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Process and Layout

Lecture 4

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SWINBURNE UNIVERSITY OF TECHNOLOGY

Acknowledgement of Country

We respectfully acknowledge the Wurundjeri People of the Kulin Nation, who are the Traditional Owners of the land on which Swinburne's Australian campuses are located in Melbourne's east and outer-east, and pay our respect to their Elders past, present and emerging.

We are honoured to recognise our connection to Wurundjeri Country, history, culture, and spirituality through these locations, and strive to ensure that we operate in a manner that respects and honours the Elders and Ancestors of these lands.

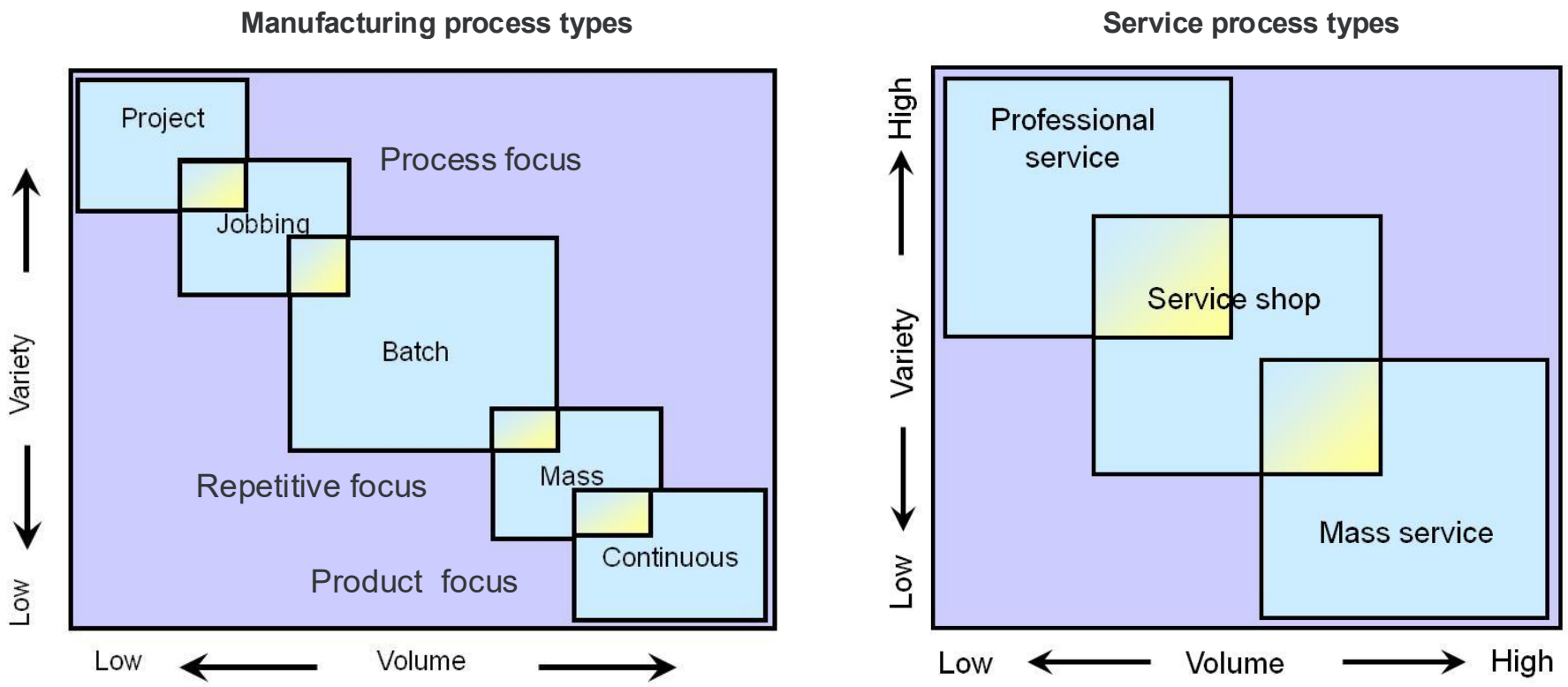
We also respectfully acknowledge Swinburne's Aboriginal and Torres Strait Islander staff, students, alumni, partners and visitors.

We also acknowledge and respect the Traditional Owners of lands across Australia, their Elders, Ancestors, cultures, and heritage, and recognise the continuing sovereignties of all Aboriginal and Torres Strait Islander Nations.

OUTLINE

- Process and layout design
- Cycle time
- Number of stages
- Layout arrangement and balancing
- Layout shape

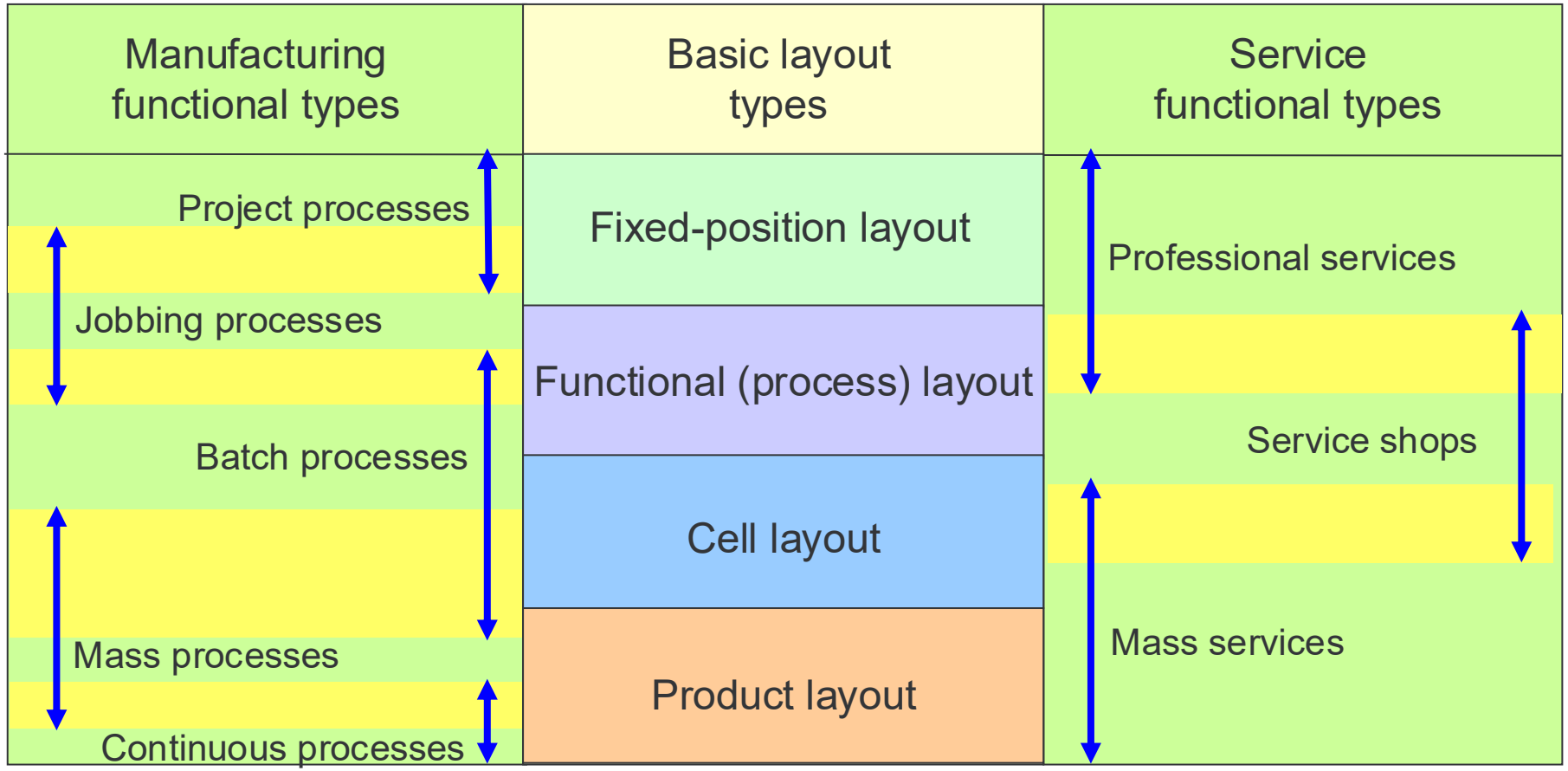
THE VOLUME-VARIETY EFFECT



Source: Slack et al. (2007)

LAYOUT AND PROCESS

- Layouts are clearly linked to processes



Source: Slack et al. (2007)

LAYOUT DESIGN: WORK CELLS AND PRODUCT - ORIENTED LAYOUTS

- Key operations principles:
  - Cycle time
  - Number of stages
  - Bottleneck principle
  - Stage variation
  - Stage arrangement
  - Process/layout shape

CYCLE TIME

- Is the time in between processed, finished, units emerge from the transformation process
- The time every finished product or serviced customer emerges from the operation
- Is this different from total processing time?

- To estimate cycle time:
  - Available time
  - Number of units to be processed

$$\text{Cycle time} = \frac{\text{Total work time available}}{\text{Units to be processed}}$$

(required to satisfy demand)

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NUMBER OF STAGES OR OPERATORS

- Benefits obtained from dividing tasks and division of labour
- Number of stages to produce or process a good or client given the total processing time (Work Content) and cycle time.
- When does the work content equals the cycle time?

- To estimate the number of stages:
  - Work content (WC): The total amount of work required to produce a unit of output
  - Cycle time (CT)

$$\text{Stages} = \frac{\text{Total processing time (WC)}}{\text{Cycle time}}$$

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EXAMPLE: COMPUTER ASSEMBLY OPERATION

Available time = 16 Hrs. (2 shifts of 8 hrs.)

No. of products to be processed = 160

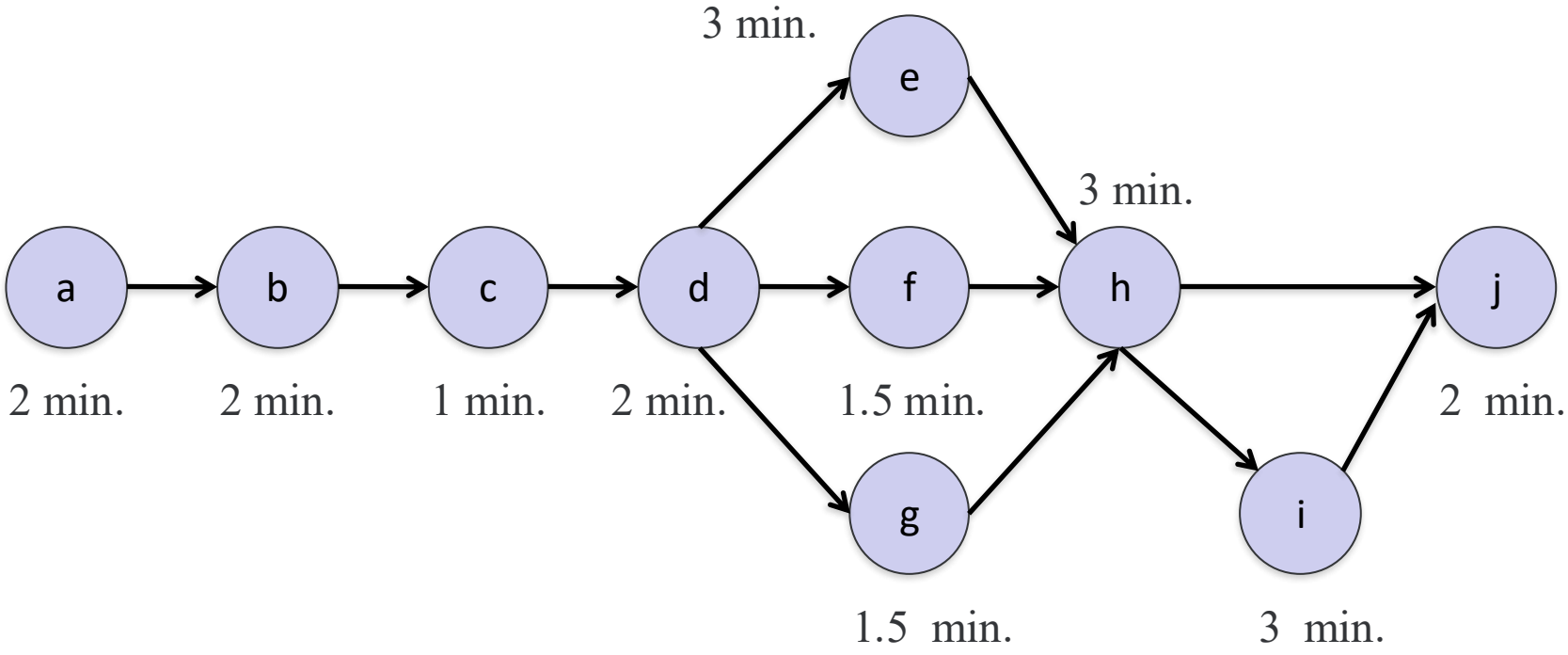
Work content = 21 min./unit

a. Motherboard assembly	=	2 min.
b. Processor assembly	=	2 min.
c. Fan assembly	=	1 min.
d. RAM memory assembly	=	2 min.
e. Video card assembly	=	3 min.
f. Audio card assembly	=	1.5 min.
g. DVD unit assembly	=	1.5 min.
h. Hard drive assembly	=	3 min.
i. Energy source assembly	=	3 min.
j. Case assembly	=	2 min.
Work content (WC)		21 min.

PROCESS TASK DETAILS: COMPUTER ASSEMBLY OPERATION

Activity code	Activity	Predecessor	Time (min.)
a	Motherboard assembly	-	2
b	Processor assembly	a	2
c	Fan assembly	b	1
d	RAM memory assembly	c	2
e	Video card assembly	d	3
f	Audio card assembly	d	1,5
g	DVD unit assembly	d	1,5
h	Hard drive assembly	e, f, g	3
i	Energy source assembly	h	3
j	Case assembly	i, h	2

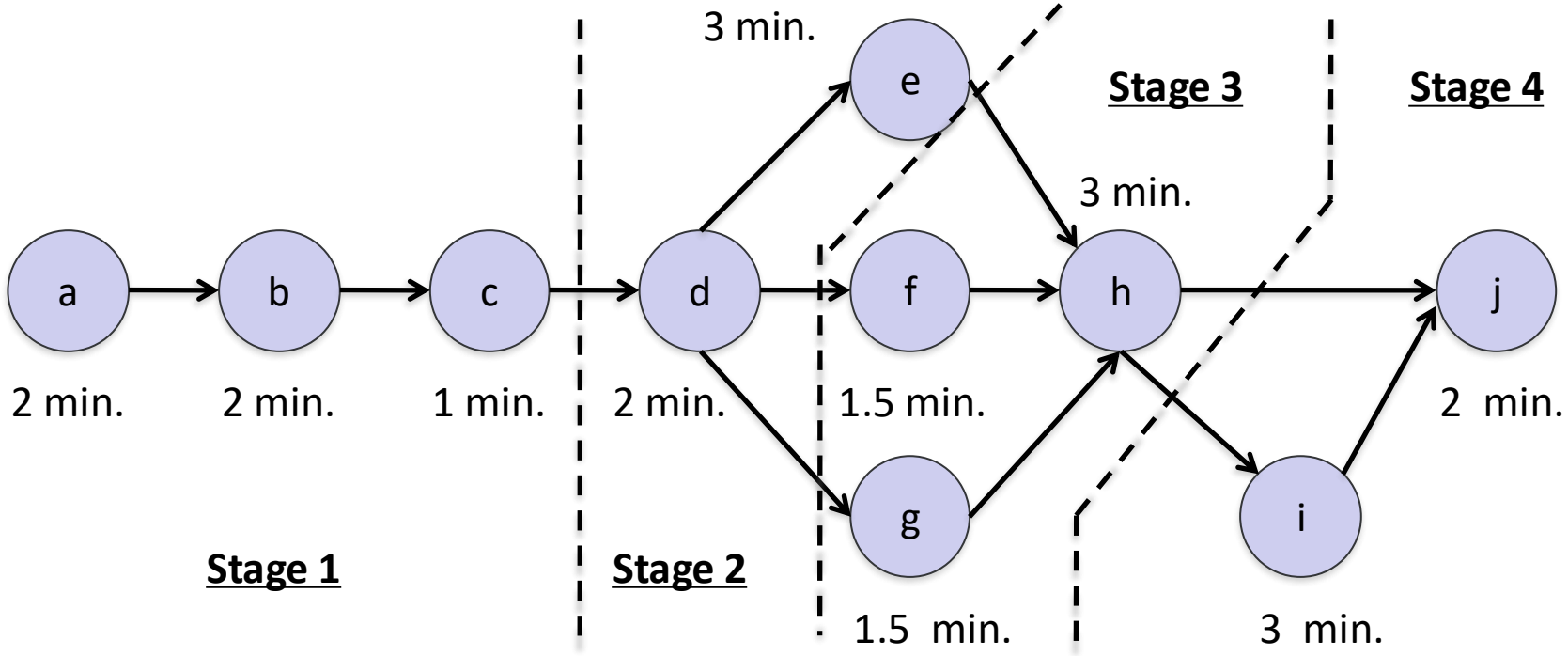
PROCESS DIAGRAM: COMPUTER ASSEMBLY OPERATION



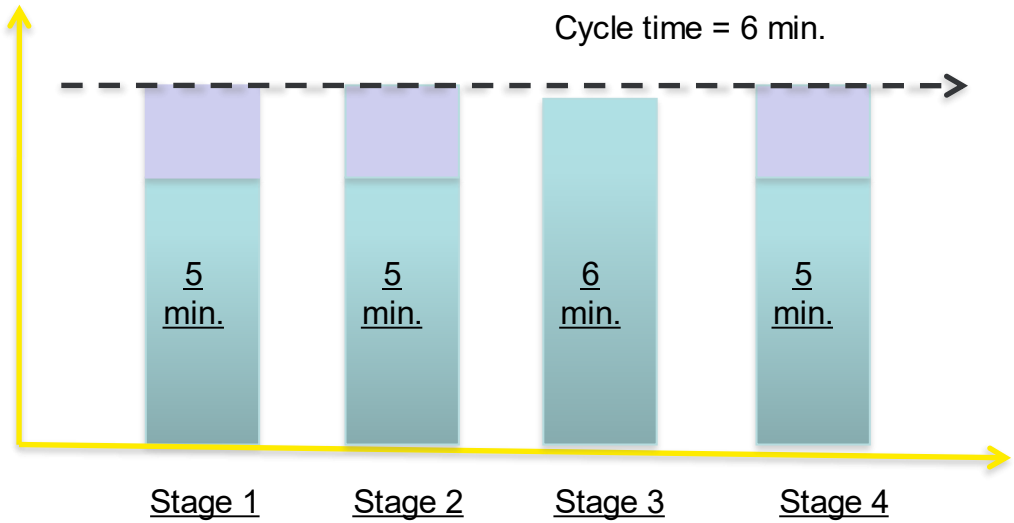
PRODUCTION STAGES

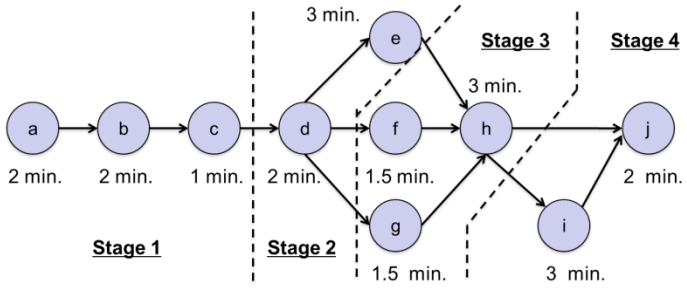
Cycle time= 16 / 160 = 0.10 Hrs. (or 6 min.)

Number of stages = 21 / 6 = 3.5 ≅ <u>4 stages</u>



LAYOUT BALANCE



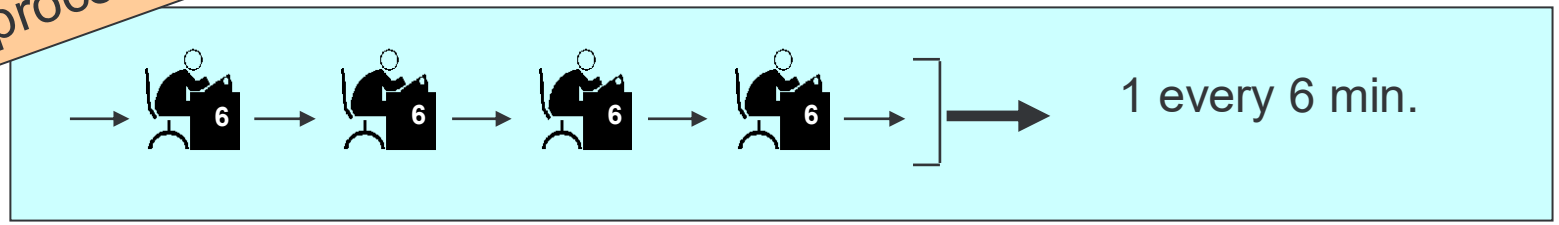


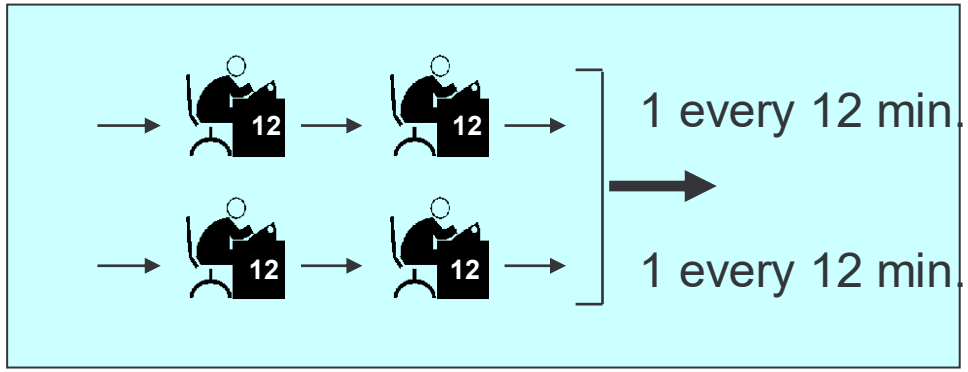
Idle cycle time = (6-5) + (6-5) + (6-5) = 3 min.

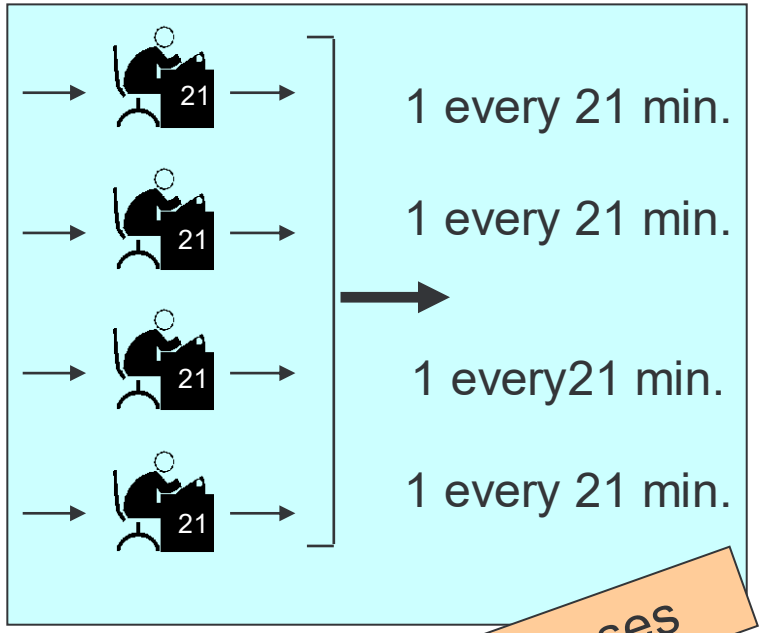
$$\text{Idle cycle time rate} = \frac{\Sigma\ (\text{cycle time - stage time})}{\text{cycle time X No. of stages}} = 0.125\ (3\ \text{min}/(6\text{X}4))\ \text{X}100$$

STAGE ARRANGEMENT

Long processes







Short processes

Advantages of long processes:

Controlled flow

Simple material handling

Lower capital requirement

Greater efficiency

Advantages of short processes:

Higher mix flexibility

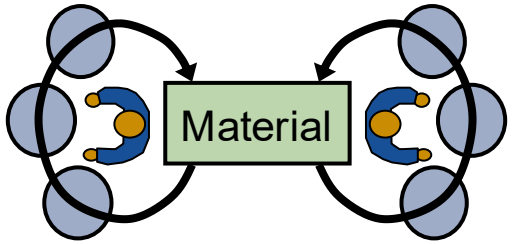
Higher volume flexibility

Less monotonous tasks

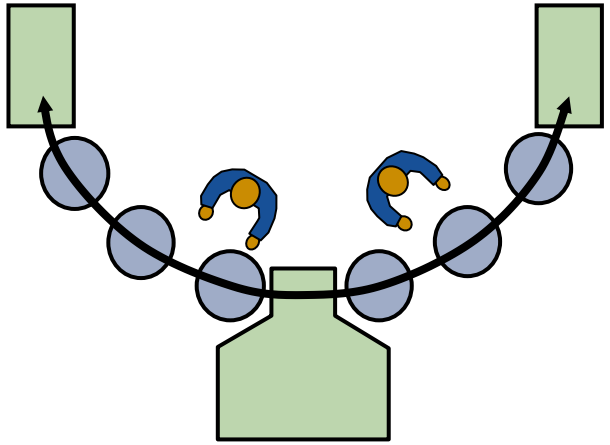
Higher ownership

LAYOUT SHAPES

Figure 9.9 (a)



Current layout - workers in small closed areas.

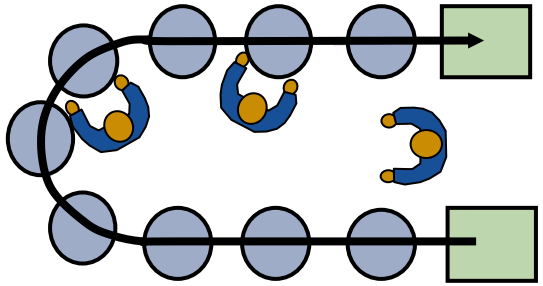


Improved layout - cross-trained workers can assist each other. May be able to add a third worker as additional output is needed.

LAYOUT SHAPES

Figure 9.9 (b)

Current layout - straight lines make it hard to balance tasks because work may not be divided evenly

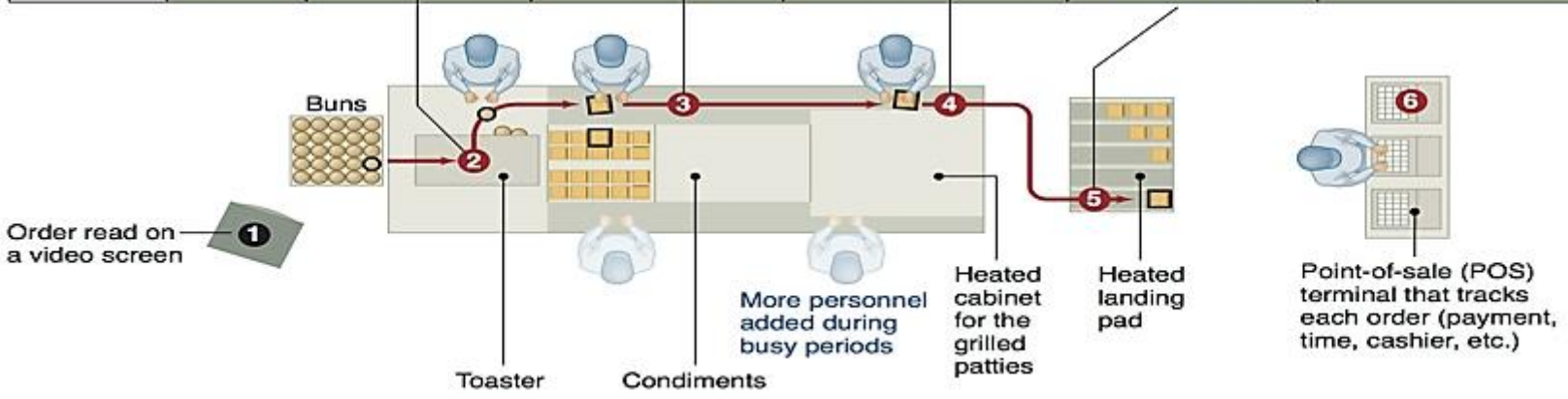


Improved layout - in U shape, workers have better access. Four cross-trained workers were reduced to three.

U-shaped line may reduce employee movement and space requirements while enhancing communication, reducing the number of workers, and facilitating inspection

ACTIVITY: MCDONALD'S ASSEMBLY LINE

Elapsed time	0:00	0:11	0:31	0:45		1:30
Task time (seconds)		11	20	14	0	45
Task	1. Order	2. Bun toasting	3. Assembly with condiments	4. Wrapping of patty with bun	5. Order picked up immediately to keep it fresh	6. Customer service (order and payment)



- What is the cycle time of this process?
- How many finished orders can we process during lunch time (2 Hrs of pick time)
- To increase the number of processed orders by 100 %, what would you do?

REFERENCES

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